

Risk-Calibrated Fallback: Certifying When an Autonomous Vehicle Should Give Up

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Abstract—Every automated vehicle has a minimum-risk manoeuvre (MRM), and the decision to invoke it is almost always a hand-tuned threshold on a confidence score — the very quantity known to become untrustworthy under distribution shift. We give the first finite-sample guarantee on *when to trigger* an MRM that is stated on the closed-loop consequence of the decision, not on the prediction behind it, and that counts the harm the fallback itself causes. We cast triggering as selective prediction and certify the probability of a missed intervention with Learn-then-Test, measured *after* the fallback has changed the future. A counterfactual rollout of every scenario prices the certificate in unnecessary stops and fallback-induced collisions, costs that open-loop evaluation cannot observe. Because a rollout’s outcome depends only on when the fallback fires, one reference run and one forced-firing rollout per decision tick evaluate every trigger, threshold and online method on identical physics. We extend the certificate to covariate shift with density-ratio weighting that exposes its effective sample size and refuses to certify when the data cannot support it. We evaluate on replayed real-world driving logs (Argoverse 2, nuScenes) under reactive traffic, with a UN R157-compliant manoeuvre and nuPlan closed-loop harm labels, against seven baselines spanning deployed practice (confidence and ensemble thresholds), open-loop conformal prediction, and online conformal decision methods. [Quantitative findings to be inserted from the pre-registered evaluation.]

Index Terms—Autonomous driving, conformal prediction, selective prediction, minimum-risk manoeuvre, closed-loop safety evaluation, trajectory prediction, risk calibration, distribution shift.

I. INTRODUCTION

AUTONOMOUS vehicles that lack confidence in their own situational understanding are expected to fall back on a minimum-risk manoeuvre: slow to a stop, pull to the shoulder, or otherwise hand control back to a human. The decision of *when* to invoke this behaviour is, in nearly every deployed and published system, a hand-tuned threshold on some scalar confidence or uncertainty score produced by a perception or prediction module. That practice has a specific and largely unexamined weakness: the very quantity the threshold is applied to — a learned model’s uncertainty estimate — is exactly what has been shown to degrade under the kind of distribution shift a vehicle experiences when it enters a new city, a new weather condition, or simply a stretch of

road unlike anything in its calibration set. A threshold tuned under one operating distribution carries no guarantee once that distribution has moved, and the failure is silent: the system continues to report a confidence number, and nothing in the number itself signals that the number is now wrong.

Conformal prediction offers a principled alternative: rather than a point confidence score, it produces a prediction region with a distribution-free, finite-sample coverage guarantee under exchangeability. A substantial and fast-growing literature has applied conformal prediction to trajectory forecasting and to planning under uncertainty in autonomous driving. Almost without exception, however, this literature is *open-loop*: it constructs a prediction region, reports its coverage and its size, and stops. Two things are invisible from that vantage point. First, a region’s effect on a downstream control decision — specifically, whether it is used to trigger a fallback — is not evaluated at all; open-loop coverage and closed-loop safety outcome are simply assumed to move together. Second, and more consequentially, *triggering the fallback is not free*. A vehicle that stops in a live lane because its uncertainty region grew large is not obviously safer than one that continues; an evaluation that reports only coverage, or only collision rate under the fallback, cannot see the trade-off between the two, and cannot see that a policy which triggers whenever it is asked to would trivially achieve a good miss rate while being useless.

This paper studies fallback triggering as a decision problem with a formal guarantee attached to the decision itself, evaluated where that guarantee actually has to hold: in closed loop, on real driving scenarios, with a trigger threshold that has consequences for what the vehicle does next. The closest prior work, Conformal Decision Theory [10], also calibrates decisions rather than predictions, but it bounds a long-run *average* risk that converges at rate $O(1/t)$, tunes a continuously acting planner parameter, and is evaluated on pedestrian navigation. An MRM trigger is different in kind. It is an irreversible, one-shot decision to abandon the task. Its errors are paid in two currencies, missed interventions and the harm of stopping itself. And a deployed vehicle needs its guarantee before the first mile, not asymptotically. We make four contributions.

1) A decision-level certificate for fallback triggering.

We formalise minimum-risk-manoeuvre triggering as selective prediction and certify, with finite-sample high-probability validity, the closed-loop *missed-intervention* rate: the probability that harm occurs without the fallback having fired first, measured after the fallback has changed the future. We use Learn-then-Test [2], which needs no monotonicity of the closed-loop loss in the threshold, and report Conformal Risk Control [3] where

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Manuscript in preparation.

monotonicity holds.

- 2) **Pricing the fallback.** Every scenario is also rolled out with the fallback disabled. This counterfactual prices each operating point in *unnecessary stops* and in *fallback-induced collisions*: collisions the ego suffers because it stopped, which nuPlan’s fault attribution correctly does not blame on the ego and which an open-loop study cannot observe at all.
- 3) **Exact counterfactual evaluation.** A rollout’s outcome depends only on *when* the fallback fires. One reference run that records every trigger score, plus one rollout forced to fire at each decision tick, therefore evaluates every trigger, threshold, online method and latency on identical physics, with no approximation.
- 4) **Certification under shift.** Density-ratio-weighted Learn-then-Test repairs the certificate under covariate shift. Its effective sample size makes explicit when the calibration data can no longer support a guarantee, so the system refuses to certify instead of silently expiring. We test it on the Argoverse 2 $\rightarrow$ nuScenes shift, where the unweighted certificate is predicted to fail.

The remainder of the paper is organised as follows. Section II situates this work against open-loop conformal prediction for driving, task-relevant failure detection, and closed-loop safety evaluation more broadly. Section III formalises the triggering problem and the calibration procedure. Section IV describes the closed-loop evaluation environment, the fallback manoeuvre, and the harm labelling used to score every rollout. Section V reports results against seven baselines, with ablations of every component. Section VI states the scope of the claims, and Section VII concludes.

II. RELATED WORK

A. Conformal prediction for autonomous driving

Conformal prediction [1] constructs prediction sets with a distribution-free, finite-sample marginal coverage guarantee under exchangeability between calibration and test data, and has been applied extensively to trajectory forecasting and to safe planning under uncertainty. Lindemann *et al.* embed conformal prediction regions directly into a model predictive controller to obtain probabilistic safety guarantees, but their guarantee collapses once the deployment distribution shifts away from the calibration distribution — a limitation Rahman *et al.* [4] address with robust conformal prediction driven by a conditional generative model of the shift, evaluated in a synthetic multi-agent corridor. SafePath [5] integrates conformal prediction into an LLM-based path-planning pipeline, providing a guarantee that a set of candidate paths contains a safe one with a user-specified probability, and reports both open-loop (nuScenes) and closed-loop (highway-env, synthetic traffic) results. In all three cases the guarantee is on the *prediction* — a region contains the truth, or a path set contains a safe path — rather than on the eventual control decision the region is used to make, and none evaluates a fallback-triggering decision on real, replayed driving logs under reactive traffic. This paper’s guarantee is instead on the decision:

the probability that a minimum-risk manoeuvre fires before a harmful event, evaluated end-to-end in closed loop.

B. Online and decision-level conformal methods

Adaptive Conformal Inference (ACI) [11] updates the mis-coverage level online so that long-run coverage holds even under arbitrary shift. Conformal Decision Theory (CDT) [10] carries the same idea from prediction sets to decisions: it updates a decision parameter $\lambda_{t+1} = \lambda_t + \eta(\varepsilon - \ell_t)$ so that the long-run average risk converges to ε at rate $O(1/t)$, with no distributional assumptions. It is demonstrated on robot navigation among pedestrians (Stanford Drone Dataset), with a continuous distance-based loss. Both give *asymptotic, average* guarantees for a quantity adjusted continuously as data arrive. We target a different object: a one-shot, irreversible trigger whose certificate must hold with finite samples before deployment, and whose errors include the harm that triggering itself causes. We adopt both methods as baselines (Section IV-E), applied episode-by-episode, which is the natural online counterpart of our setting.

C. Task-relevant and closed-loop failure detection

Farid *et al.* [6] propose a run-time monitor for trajectory-prediction failures that is *task-relevant* by construction: it flags a misprediction only when the resulting planning cost is anomalous relative to the predicted cost distribution, with data-free false-positive and false-negative rate bounds. Their detector is evaluated against hand-labelled scenarios in an open-loop and semi-reactive setting, and — like the conformal-prediction literature above — does not close the loop through an actuated fallback and re-measure the outcome. Their central insight, that a task-irrelevant misprediction should not count against a detector, is reflected in this paper’s harm labelling: a collision the ego vehicle suffers while already stopped or while being struck from behind is scored as a distinct *induced-collision* cost of the fallback itself, never as a missed intervention.

D. Risk-sensitive safety analysis

Chapman *et al.* [7] develop risk-sensitive safe sets using Conditional Value-at-Risk to assess the severity, not merely the probability, of constraint violation along a stochastic trajectory, providing the risk-sensitive vocabulary this paper borrows in motivating why a mean missed-intervention rate is the appropriate quantity to control at a chosen operating point, even though a tail-severity extension (Section VI) is left to future work.

E. Closed-loop simulation infrastructure

ScenarioNet [8] unifies heterogeneous real-world driving logs — including Argoverse 2, nuScenes, nuPlan, and Lyft Level 5 — into a common scenario description that can be replayed interactively in the MetaDrive simulator, with either logged-trajectory replay or reactive rule-based (IDM) traffic. This work is built directly on that infrastructure: every scenario reported in Section V is a real logged driving scenario, replayed headless with reactive traffic, not a synthetic or hand-crafted one.

III. PROBLEM FORMULATION AND METHOD

A. Fallback triggering as selective prediction

Consider a closed-loop rollout of a scenario X under a fallback policy governed by a scalar threshold λ : at every decision tick the vehicle computes an uncertainty score $u_t(X)$ from a trajectory predictor's output for nearby agents, and engages the minimum-risk manoeuvre the first time $u_t(X) > \lambda$, remaining engaged for the rest of the episode. Define the closed-loop loss

$$L(X, \lambda) = \mathbb{I}[\text{harm in } X \text{ under } \lambda, \text{ MRM not first}], \quad (1)$$

the *missed-intervention* indicator, and the complementary over-conservatism quantity

$$S(X, \lambda) = \mathbb{I}[\text{MRM fires under } \lambda, X \text{ harm-free at } \lambda = \infty], \quad (2)$$

the *unnecessary stop* indicator, evaluated against the counterfactual rollout of the same scenario with the trigger disabled. The goal is to choose $\hat{\lambda}$ such that

$$\Pr_{X_1, \dots, X_n} \left(\mathbb{E}_X [L(X, \hat{\lambda})] \leq \alpha \right) \geq 1 - \delta, \quad (3)$$

while minimising $\mathbb{E}_X [S(X, \hat{\lambda})]$ — the smallest autonomy loss compatible with the risk target α .

Equation (3) is not a conformal coverage statement: $L(X, \lambda)$ depends on the entire closed-loop rollout under λ , since engaging the fallback changes every subsequent state, so it is neither a marginal coverage event on a static prediction nor guaranteed to be monotone in λ a priori. We therefore calibrate $\hat{\lambda}$ with Learn-then-Test [2]: for a finite grid $\lambda_1 > \lambda_2 > \dots > \lambda_J$ ordered from least to most conservative, compute the empirical risk $\hat{R}(\lambda_j) = \frac{1}{n} \sum_i L(X_i, \lambda_j)$ on n calibration scenarios, a Hoeffding–Bentkus p -value for the null hypothesis $H_j : \mathbb{E}[L(\cdot, \lambda_j)] > \alpha$, and apply fixed-sequence testing starting from the most conservative λ ; every λ_j whose null is rejected satisfies (3) simultaneously, with family-wise error controlled at δ , and $\hat{\lambda}$ is the least conservative such λ_j (largest autonomy). Where the empirical loss is observed to be monotone in λ across the calibration set, we additionally report Conformal Risk Control [3], whose expectation guarantee is tighter but requires that monotonicity assumption; where it is violated we report the violation rate directly rather than silently falling back to the weaker method.

B. Trigger score

At each decision tick the trigger score u_t measures how far a predicted future position of any nearby agent intrudes into the ego vehicle's own near-term extent. Writing the ego's constant-heading, constant-speed extrapolation over a horizon of T steps as $\{p_\tau^{\text{ego}}\}_{\tau=1}^T$ — the same route-agnostic motion model used by the harm metric in Section IV-B, deliberately *not* the vehicle's planned, route-following path, so that the trigger cannot credit itself for steering behaviour the safety label does not assume — and a predictor's K -mode forecast for agent i as $\{\hat{p}_\tau^{(i,k)}\}$, each predicted offset is decomposed into a longitudinal component $a_\tau^{(i,k)}$ (along the ego heading) and a lateral component $\ell_\tau^{(i,k)}$ (perpendicular to it), each compared against a separate margin: a length-based longitudinal margin

$m_i^{\text{lon}} = \frac{1}{2}(L_{\text{ego}} + L_i) + c_{\text{lon}}$ and a width-based lateral margin $m_i^{\text{lat}} = \frac{1}{2}(W_{\text{ego}} + W_i) + c_{\text{lat}}$. The intrusion at (i, k, τ) is

$$\rho_\tau^{(i,k)} = \max \left(0, 1 - \sqrt{\left(\frac{a_\tau^{(i,k)}}{m_i^{\text{lon}}} \right)^2 + \left(\frac{\ell_\tau^{(i,k)}}{m_i^{\text{lat}}} \right)^2} \right) m_i^{\text{lat}}, \quad (4)$$

and $u_t = \max_{i,k,\tau} \rho_\tau^{(i,k)}$. A single isotropic clearance radius cannot represent both margins without either missing a straight-ahead lead vehicle (a radius set by width alone) or over-triggering on adjacent-lane traffic (a radius set by the vehicle's half-diagonal); the decomposition in (4) is required for correctness, not merely convenient, a fact confirmed empirically during development (Section IV).

C. Fallback manoeuvre

On triggering, the vehicle executes the minimum-risk manoeuvre defined by UN Regulation No. 157 (Automated Lane Keeping Systems) § 5.5.1 [16]: decelerate inside the current lane with a deceleration demand not exceeding 4.0 m/s^2 , lane-keeping control otherwise unchanged, holding the vehicle stationary once stopped. This is enforced by a proportional controller tracking the target speed profile $v(t) = \max(v_0 - 4.0t, 0)$ from the moment of triggering.

D. Certification under covariate shift

The certificate in (3) assumes calibration and deployment scenarios are exchangeable. When the deployment distribution differs by covariate shift, we reweight calibration scenarios by an estimated density ratio $w(x) \propto p_{\text{target}}(x)/p_{\text{source}}(x)$ computed from pre-deployment scenario covariates only (agent counts and speeds, initial gap, route length). A probabilistic domain classifier is trained on unlabelled target scenarios, following weighted conformal prediction [12]. The weighted empirical risk $\hat{R}_w(\lambda) = \sum_i w_i L(X_i, \lambda) / \sum_i w_i$ is tested with the Hoeffding–Bentkus p -value at the Kish effective sample size $n_{\text{eff}} = (\sum_i w_i)^2 / \sum_i w_i^2$. As the weights concentrate, n_{eff} shrinks and the test certifies a more conservative threshold, and eventually none at all. The procedure thus reports when the source data cannot support a guarantee for the target, rather than asserting one that has silently expired.

E. Exact counterfactual evaluation

Two rollouts of the same scenario are identical until one of them triggers the fallback, up to the simulator's own non-determinism. We measure this in every forced-firing rollout. Over the first 33,761 forced-firing rollouts (2,000 scenarios), 98.4% agreed with the reference run to within 1 cm before firing. The pre-fire time-to-collision trace differed in 1.2% and the pre-fire harm status in one rollout (0.003%). We report the result of every analysis both on all scenarios and after dropping the 3.6% of scenarios with any flagged drift. A rollout's outcome is therefore a function of the *firing tick* alone, not of which score, threshold or update rule caused the firing. For each scenario we simulate one reference run with the fallback disabled, recording every trigger score at every decision tick,

and one rollout forced to fire at each decision tick. Any rule of the form “fire at the first tick where $u_t > \lambda$ ” is then an exact lookup. This includes online rules, whose λ changes between scenarios, and added actuation latency. All methods in Section V are compared on the same simulated physics, so differences between them cannot come from simulator noise or from unequal compute budgets.

IV. EXPERIMENTAL SETUP

A. Simulation environment

All rollouts are executed headless in MetaDrive, replaying scenarios converted by ScenarioNet [8] from Argoverse 2 Motion Forecasting and nuScenes. Non-ego traffic is controlled reactively (rule-based, IDM), not replayed open-loop from the log: pure log replay was found to roughly triple the no-fallback collision rate on a 1,000-scenario diagnostic sample, because logged agents cannot react to an ego vehicle that deviates from its logged trajectory — precisely the deviation a fallback manoeuvre produces. Reporting results under non-reactive replay would therefore misrepresent the cost of stopping, which is a central quantity this paper reports.

B. Harm labelling

A rollout is labelled harmful using the nuPlan closed-loop safety definition [9]: an at-fault collision, or a time-to-collision below 0.95 s computed over a 3.0 s horizon at 0.1 s resolution, not evaluated while the ego is stationary and not evaluated against tracks behind the ego. Fault attribution follows the nuPlan convention: a collision the ego suffers while stationary, or while being struck from behind, is *not* at-fault and is never counted as a missed intervention; it is instead recorded as an *induced collision* — the concrete, measurable cost of a false or premature trigger.

C. Predictor

The trajectory predictor is AutoBot [14] (6 modes, 2.1 s history, 6 s horizon), trained with the UniTraj framework [15] on the Argoverse 2 Motion Forecasting training split (minADE₆ = 1.09 m on held-out AV2 validation scenes). It runs online inside the closed loop, so its input is the actual, possibly ego-deviated, simulator history rather than the original log. This is necessary once traffic is reactive. The model’s own UniTraj preprocessing is reused unchanged, so it sees its training input distribution. Vehicles are predicted by the network. Pedestrians and cyclists, which the checkpoint was not trained on, receive constant-velocity forecasts.

D. Datasets and splits

All scenarios come from Argoverse 2 validation, disjoint from the predictor’s training data. They are split deterministically by scenario identifier into development (14,990), calibration (4,971) and test (5,027) sets. The shifted target is nuScenes (9,041 scenarios of 8.1 s, Boston and Singapore, versus Argoverse 2’s six U.S. cities). Scenarios in which the ego moves less than 10 m are excluded, as in ScenarioNet. Design choices were made on the development split only.

The pre-registered evaluation uses 2,000 calibration, 2,000 test and 1,500 nuScenes scenarios. Decisions are taken at 2 Hz. A paired development-split comparison showed identical miss rates at 2 Hz and 10 Hz, and a 10 Hz replication on test scenarios is reported.

E. Baselines

All methods use the same predictor, the same MRM and the same test scenarios, and are evaluated exactly (Section III-E).

- **T1 — confidence threshold** (deployed practice): fire when $1 - \max_k p_k$ of any nearby vehicle exceeds a threshold. The threshold is tuned on calibration data to the most permissive value with empirical miss rate $\leq \alpha$.
- **T2 — ensemble threshold**: the same tuning, applied to the spread of endpoint predictions across five Monte Carlo dropout passes [13].
- **T3 — open-loop conformal**: inflate every predicted mode by the split-conformal radius of the predictor’s 3 s error on held-out Argoverse 2 logs at level α [1], and fire when the inflated region touches the ego. This is the guarantee-on-the-prediction design that the paper argues is insufficient.
- **Tuned geometric**: our score with an empirically tuned threshold and no certificate. It isolates the contribution of the calibrator.
- **CDT** [10] and **ACI** [11]: online decision- and prediction-level adaptation, updated after every scenario. Each uses the step size that performs best on test, which favours the baselines.
- **Oracle**: the best threshold in hindsight on test data. It is not deployable and is reported as an upper bound.

F. Metrics and statistics

For every method we report the missed-intervention, unnecessary-stop and fallback-induced-collision rates, autonomy (the fraction of scenarios completed without the fallback) and route completion, with Wilson 95 % intervals. Validity of each certificate is assessed over 200 random resplits of the pooled calibration and test scenarios. A valid certificate keeps the frequency of test miss rate above α at or below δ . Every conclusion is also reported under six alternative harm definitions.

G. Pre-registration and reproducibility

The harm definition, $\alpha = 0.05$, $\delta = 0.10$, and the refutation criteria for all three hypotheses were recorded in a version-controlled note before any calibration or test data were collected. The note also records its dated amendment and everything observed on the development split beforehand. Code, per-scenario rollout records, configuration files and the note itself are released with the paper. Each scenario’s record contains every score trace and outcome, so every table can be regenerated without re-simulation.

V. RESULTS

Results in this section are pending the full-scale calibration and test-split evaluation ($n \approx 2,000$ scenarios per split) and are not yet reported. Diagnostic runs at small scale ($n \leq 300$) were used exclusively to validate the pipeline described in Sections III–IV — in particular, to identify and correct two errors in the trigger-score geometry of (4) before any pre-registered evaluation — and are not reported here as results.

A. Main comparison (H2)

[Pending: Table — miss rate, unnecessary stops, induced collisions, autonomy for T1, T2, T3, tuned geometric, CDT, ACI, LTT, CRC and oracle on Argoverse 2 test. Figure — miss rate versus unnecessary-stop rate, every method at its operating point, oracle frontier overlaid.]

B. Is the certificate valid?

[Pending: distribution of realised test miss rates over 200 resplits per method, with violation frequency against δ .]

C. Open-loop versus closed-loop ranking (H1)

[Pending: AUROC ranking versus certified unnecessary-stop ranking of six scores; Kendall's τ .]

D. Dataset shift and its repair (H3)

[Pending: Argoverse 2-certified threshold deployed on nuScenes, unweighted versus weighted Learn-then-Test, with n_{eff} .]

E. Ablation study

[Pending: each component removed in turn, all certified by Learn-then-Test on identical scenarios. Score components: route-following instead of kinematic corridor; width-only isotropic radius instead of the length-aware box; confidence or ensemble score instead of geometry. Calibrator: Learn-then-Test versus Conformal Risk Control versus empirical tuning. Actuation latency of 0, 0.5 and 1.0 s.]

F. Sensitivity

[Pending: non-reactive log-replay traffic; a 2.0 m/s^2 comfort MRM; 10 Hz decisions; six harm definitions.]

VI. SCOPE AND FUTURE WORK

The claims above are deliberately scoped. Each boundary below is a design choice with a stated rationale, and each suggests a direct extension.

Replayed real scenarios, simulated responses. Every scenario is a real logged traffic situation, but agents respond to the ego through a rule-based model. We bracket this choice by reporting non-reactive replay alongside reactive traffic, and we treat as robust only the conclusions that hold under both. Learned, data-driven agent models are a natural next step, and the evaluation protocol accepts them unchanged.

Marginal, scenario-level guarantees. The certificate bounds the miss rate over the scenario distribution, the quantity a deployment authority can audit. It does not bound the risk of any individual scenario. Stratified certification by scenario type follows directly from the same procedure applied within strata.

Covariate shift. The weighted certificate covers shift in observable scenario covariates. Shift in the conditional behaviour of road users given those covariates is outside its guarantee. The effective sample size exposes when the weights, and hence the certificate, become unreliable.

Object states as input. The predictor receives agent states, which isolates the triggering decision from perception error. Composing the certificate with a calibrated perception module is future work that this paper's framework is designed to accept.

A mean-risk certificate. We bound the probability of a missed intervention. The same procedure bounds any bounded loss, so a severity-weighted or tail (CVaR) certificate in the sense of [7] requires only a change of loss.

One fallback family. We evaluate stop-in-lane manoeuvres at two deceleration levels, as specified by UN R157. Other fallbacks, such as a lateral pull-over or a hand-over, change the costs being priced but not the method.

VII. CONCLUSION

[To be completed once Section V is populated.]

ACKNOWLEDGMENT

The authors thank the maintainers of MetaDrive and ScenarioNet for releasing the simulation infrastructure this work builds on.

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